Venkata Lakshmi Parimala Pasupuleti

Full Stack Engineer · Regulatory Compliance AI · Agentic Workflow Automation · React / TypeScript / PostgreSQL · 9yr Production

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PROFESSIONAL SUMMARY

Full stack engineer specializing in AI-powered regulatory compliance automation -- I built LeaseI, an 8-agent LangGraph platform that extracts regulatory rules from complex compliance corpora, applies business logic across agentic workflow stages, and routes edge cases to human reviewers. That is the same architecture PermitFlow applies to construction permitting: document intelligence, regulatory rule enforcement, agentic routing, and HITL escalation for decisions that carry legal consequences. At Entrata, I owned HUD/LIHTC/HCV compliance automation for 7 years -- annual ingestion of federal income-limit data, eligibility calculations, recertification workflows for 100K+ housing units, zero compliance violations. I bring full-stack depth (Python/FastAPI backend, React/TypeScript/Angular frontend, PostgreSQL), production AI systems (LangGraph, LLM APIs, CI quality gates), and direct experience building the exact software PermitFlow is scaling. Portfolio: venkatalakshmiparimala.github.io/Portfolio/about.html

TECHNICAL SKILLS

Languages & Frameworks: Python (9yr production), TypeScript, JavaScript, SQL/PL-SQL, R, Bash/shell scripting, HTML/CSS; FastAPI, Flask, Angular, React.js, Django (familiarity), Node.js (ramping)

AI, Compliance Automation & Data: LangGraph (agentic orchestration), LangChain, Agentic AI, Multi-Agent Systems, LLM APIs (Anthropic, OpenAI, Groq), RAG Pipelines, Document Intelligence, OCR, Regulatory Compliance Automation, Permitting Workflows, HITL, LLM Evaluation, LLM-as-Judge, Hallucination Detection, Guardrails, Embeddings, NLP, Prompt Engineering, Machine Learning (Random Forest, Isolation Forest, XGBoost), Pandas, NumPy, scikit-learn

Backend, Data & Infrastructure: PostgreSQL (200+ queries optimized, stored procedures, multi-tenant schemas), MySQL, Oracle, NoSQL, ETL/ELT Pipelines, Data Modeling, REST APIs, Microservices, AWS (S3, EC2, RDS, Glue), Docker, Kubernetes, RabbitMQ async processing, CI/CD (GitHub Actions), Automated Testing, Regression Testing, Observability, Git, Agile/Scrum, Code Reviews

EXPERIENCE

Entrata -- Multifamily Real Estate SaaS (PropTech)

Sep 2018 -- Aug 2024

Senior Software Engineer -- Regulatory Compliance Automation, Full Stack & Data Pipelines · Lehi, UT (Remote)

- **7-year ownership of HUD/LIHTC/HCV regulatory compliance automation** -- sole owner of Entrata's federal income-limit import and eligibility system: annual ingestion of complex federal regulatory data (few thousand source rows expanded to ~180,000 domain records), AMI threshold enforcement, household income certification, asset documentation, and recertification workflows for 100K+ housing units. Every release required pre-import validation, reconciliation, and rollback preparation -- a wrong import means thousands of residents get the wrong compliance decision. Zero compliance violations across full ownership period. Direct analog to PermitFlow's permit accuracy and compliance SLA.
- **Full stack feature delivery end-to-end** -- shipped 20+ full-stack production modules (Python, Angular/JavaScript, PostgreSQL) across 120+ Agile sprints; 300+ peer code reviews; Docker microservices redesign (50% scalability improvement, 30% deployment time reduction); eliminated 80% of manual compliance-operations workload through Python/Bash automation pipelines. Partnered directly with Ops and Customer Success to translate compliance edge cases into software rules.
- **Backend and data pipelines at scale** -- RabbitMQ async processing with idempotency guarantees; optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, and partitioning; complex schemas for eligibility, financial, and document workflows across multi-tenant environments; AWS S3 migration with Libsodium-encrypted ETL pipelines (100% PHI compliance, zero security incidents).
- **Mentorship and engineering culture** -- mentored 5+ engineers; built internal debugging tools reducing debugging time by 40%; kept bug count below 10 for 6+ consecutive months at 99.5% uptime.

George Mason University

Aug 2024 -- May 2026

Research Associate & Graduate Teaching Assistant

- Built multi-source ML prediction pipelines (Python, R) across 190+ countries (R^2=0.977); applied Isolation Forest to automate data integrity checks across 3 research pipelines; designed Tableau and ggplot2 dashboards improving research comprehension 40% for faculty leadership; mentored 600+ students in Python, SQL, and data engineering.

Magizhs Tech Solution

May 2025 -- Aug 2025

AI/ML Engineer -- Agentic Compliance & Regulatory Workflow Automation

- Built an agentic compliance system (LangGraph stateful graph: parse → plan → retrieve → validate → conditional routing) with RAG over FAA Part 107 regulatory corpus -- every compliance check grounded in cited regulation; automated validation guardrails enforcing altitude limits, battery budgets, and geometric no-fly-zone checks before export. Same pattern as PermitFlow's AI agents navigating permit regulations.

Entrata India -- Pune

Jan 2017 -- Sep 2018

Associate Software Engineer -- Python · PostgreSQL · Compliance Workflows

- Built compliance and property management modules (Python, PostgreSQL, Angular); income-limit import pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints; 56 state-specific housing certification templates.

PROJECTS

Leasel: AI-Powered Regulatory Compliance Platform -- Direct PermitFlow Architecture Analog

LangGraph · Compliance AI · Agentic Workflows · HITL · FastAPI · Document Intelligence

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · SQLite · OCR · pdfplumber · GitHub Actions | github.com/venkatalakshmiparimala/Lease_complaince_rag

AI compliance platform for real estate regulations -- the same architecture PermitFlow scales for construction permits. **Agentic regulatory workflow:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → HITL Escalation) with multi-surface shared evidence model; 93.6% precision and 87.9% F1 across 1,240 compliance test cases on real estate regulatory documents (5 US states + federal, 1,050 synthetic cases). **Document intelligence pipeline:** OCR-fallback ingestion, section-aware chunking, TF-IDF retrieval, 100% citation correctness, missing/conflicting clause detection. **Core design rule enforced in code:** LLM extracts and reasons; human or deterministic logic makes every final regulated compliance call -- directly applicable to permit decision workflows. **Quality gates:** CI regression gate blocking deploys on score degradation; LLM-as-judge scoring; multi-trial sampling.

AgentProbe: Open-Source AI Evaluation Toolkit -- Production Quality Gates for AI Workflows

AI Evaluation · CI/CD · LLM APIs · FastAPI · React · TypeScript

Python · FastAPI · React.js · TypeScript · ChromaDB · GitHub Actions · Anthropic / OpenAI / Groq APIs | github.com/venkatalakshmiparimala/AgentProbe

Production AI evaluation framework ensuring AI-driven workflows ship only when quality thresholds are met. 5-metric automated evaluation engine (correctness, groundedness, latency, token efficiency, hallucination rate); CI-integrated regression gate (GitHub Actions) blocking quality degradation; multi-provider LLM routing (Anthropic, OpenAI, Groq); 15-endpoint FastAPI backend; React.js + TypeScript frontend. Used to evaluate LeaseI compliance AI across 1,240+ test cases. Demonstrates the same quality discipline PermitFlow's compliance-critical AI workflows require.

AI on FHIR: Full Stack Flask + React Compliance Platform

Flask · React · TypeScript · LLMs · Full Stack · HIPAA Compliance

Python · Flask · React · Docker · spaCy · BERT · Embeddings · FHIR API | github.com/venkatalakshmiparimala/AI_on_FHIR

HIPAA-compliant Flask + React full-stack compliance platform translating natural-language queries into regulated API calls using LLMs, BERT, and embeddings -- reduced manual retrieval time ~60%; enforced RBAC across structured medical records. Demonstrates full-stack delivery (Flask backend + React frontend) on regulated compliance data.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2013--2017